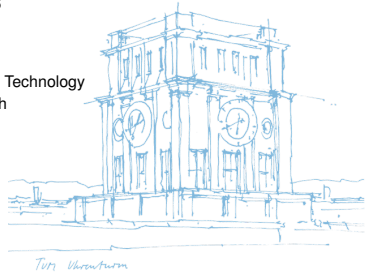


An Overlay for Social Networks based on Directed Acyclic Graphs on the Edge

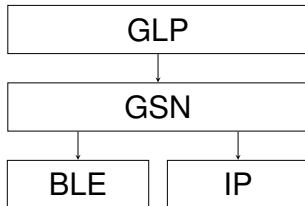
Dan Bachar, David Guzman

Wednesday 22nd April, 2026

Chair of Connected Mobility
School of Computation, Information, and Technology
Technical University of Munich

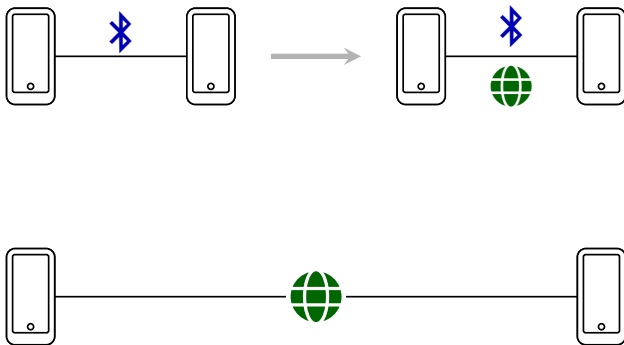


Architecture



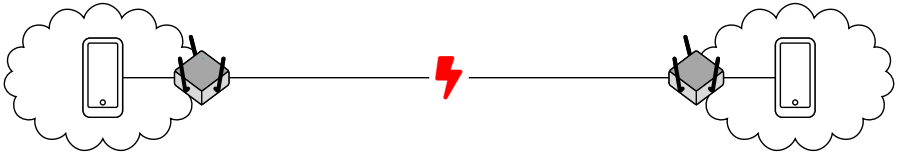
- Connect two peer devices according to priority and proximity pair
- Dual-stack network comms: BLE first, UDP (UDX) second
- Use BLE for signaling
- ICE-similar stack for address reflection, hole-punching

Dual-stack comms

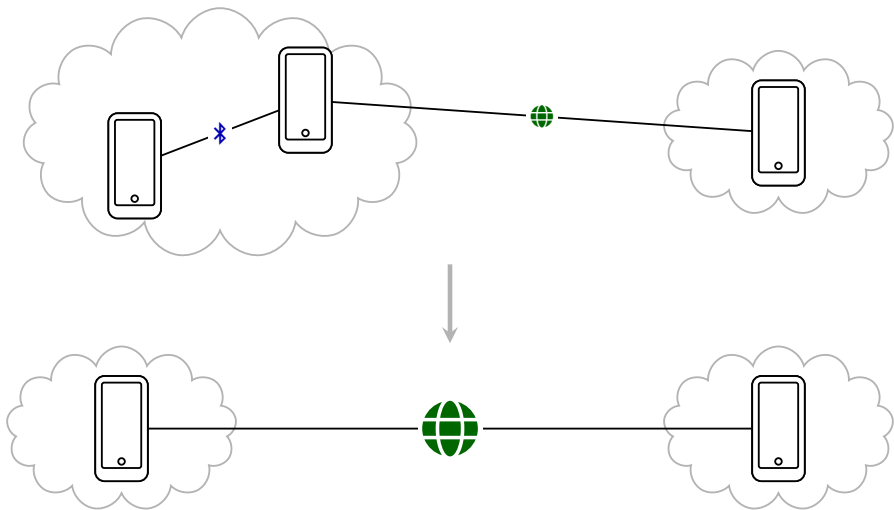


- BLE as signaling channel
- Advertise multiple addresses
- UDP for hole-punching
- UDX for long-range communication

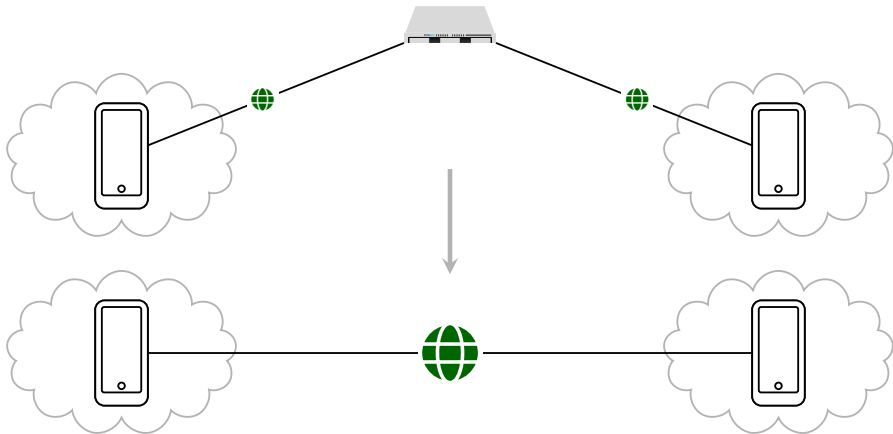
NAT



Well-connected friends



Rendezvous Server



Next steps

- Stabilize connectivity, testing
- Deploy GLP compiled to dart code on smartphones
- Write